

Leaching is a preferential solution of one or more constituents of a solid mixture by contact with a liquid solvent.

Leaching. Originally referred to percolation of the liquid through a fixed bed of solids.

Lixivation is a less frequently used synonym of Leaching.

↳ It originally referred to leaching of alkali from wood ashes.

Decoction: It is a special case of leaching when where the solvent is at boiling temperature.

Elutriation or Elution:- When the soluble material is largely on the surface of an insoluble solid and is merely washed off by the solvent.

Leaching is most Extensively used in the Metallurgical Industry.

Many other chemicals also are separated by leaching: examples include sugar from sugar beet by leaching with hot water, vegetable oil from seeds like soyabean and cottonseed with organic solvent.

* Tea and Coffee are prepared by leaching.

- * Often the solid has to be given a pre-treatment. Crushing and grinding are preferred in this regard, that enhances leaching. This is a preferred route in metallurgical industry.
 - * For extraction of product from vegetable and animal bodies, which are cellular in str. The product to be leached in most cases remains within the cell walls. Which is to be leached out of the cell wall - Thus leaching involves osmotic passage thru the cell wall. It is however not desirable to crush everything, as that would bring undesirable albumin and colloids to the product.
- Thus for extracting sugar from beets, the ~~sugar~~ sugar beets are ~~are~~ cut into small wedge-shaped slices called cossettes. This reduces the diffusion path.

Leaching Temperature:

It is generally desirable to do a leaching at as high temperature as possible, as at higher temperature the solubility is maximum. Also, ~~the temp~~ at higher temperature, the viscosity is ~~is~~ lower and diffusivity is higher.

Solution Mining: - In Place (in situ) Leaching.

This refers to percolation leaching of minerals in place at the time, by the circulation of solvent, over and thru' the ore body.

Calculation:-

It is for estimating the extent of leaching which can be obtained for a given procedure, that is to calculate the amount of soluble substance leached from a solid, knowing the initial solute content of the solid.

Stage Efficiency

Solid is leached with more than enough solvent to dissolve all the soluble solute present. If further there is no preferential adsorption of either solvent or solute by the solid,

If adequate time of contact of solid and solvent is permitted, then all the solute will dissolve in the solvent

The ~~insoluble~~ insoluble solid is then separated from the liquid, ~~by~~ physically drainage, / settling / filtration etc. The entire operation constitutes one stage,

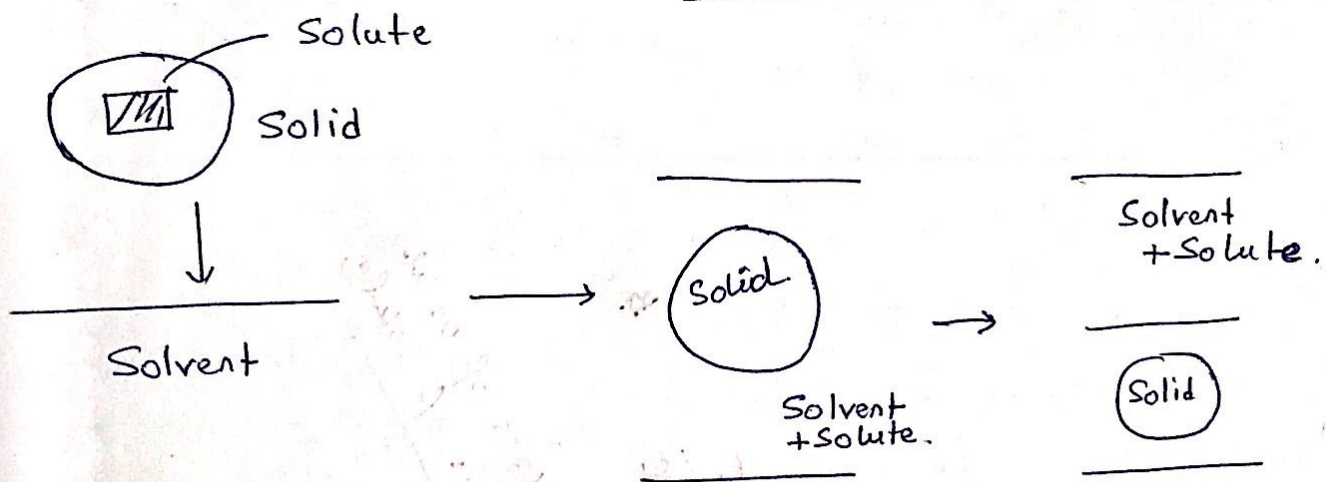
This would be an equlbm. stage with 100% stage efficiency.

In practice, stage efficiencies are much less than this,

because (1) the solute may not be completely dissolved, due to inadequate contact time, (2)

The liq solid mech sepⁿ with 100% efficiency is impractical; and solid leaving the stage ~~will~~ will always retain some liquid and its associated dissolved solute.

In case solute is adsorbed ~~on~~ ^{on} liquid ~~by~~ the solid, even though equilibrium between the liq and solid phase is obtained, imperfect settling or drainage will result in lower efficiency.



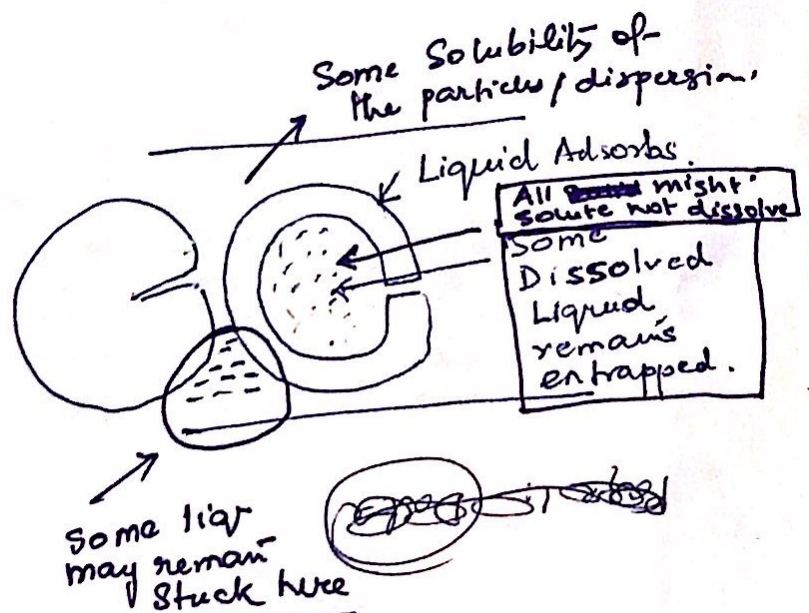
Inefficiency can be due to

- ① Solute ~~at~~ adhering on to the Solid.
- ② Solution might be entrapped. in the solid.
- ③ Some solid might be dissolved in the solvent. / dispersed (Brownian motion, if solid particles too small),
- ④ ~~Not~~ All the solute might not dissolve from the ~~solid~~ solid, to the solvent.

Time requirement is also an important factor.

Some liquid can

- (1) Remain stuck inside the pores,
- (2) Outside between the particles,
- (3) Adsorb on the surface of the particles,



Solid to be Leached

B, Mass of insoluble.

F, Mass of A+C.

$$N_F = \frac{B}{A+C}$$

$$y_F = \frac{C}{A+C}$$

Leached Solid (E_1)

B, Mass of Insoluble.

E_1 = mass of (A+C)

$$N_1 = \frac{B}{E_1} = \frac{B}{(A+C)E_1}$$

$$y_1 = \frac{C \bar{u} E_1}{(A+C)E_1} = \frac{C \bar{u} E_1}{E_1}$$

Leached Solution (R_1)

R_1 = Mass of Soln.

$$x_1 = \frac{\text{mass of C}}{\text{mass of (A+C)}}$$

Leaching Solvent

R_0 , mass of Soln. (A+C)

$$x_0 = \frac{C}{A+C}$$

$$N_0 \approx 0$$

In leaching,

A → Solvent

B → Insoluble

C → Solute

*

In Case the ~~Solution~~, no adsorption of Solute occurs and, so that the withdrawn ~~solid~~, Solution and Solution, associated with the Solid, have same composition, and the tie lines becomes vertical,

The ~~equilibrium~~ equilibrium curve is a 45° line and, distribution coefficient is 1.

For a feed that is dry,

N = Ratio of Insoluble to Soluble Substance = $\frac{B}{C}$

And $y_F = 1.0$. ($y_F = \frac{C}{A+C}$, If dry $A=0$)

And for a pure Solvent (pure A)

$$N=0$$

$$\underline{x=0}$$

Concept of Practical Equilibrium

Solid to be Leached.

B, Mass of Insoluble

F, Mass of A+C.

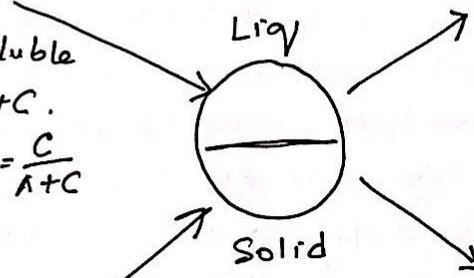
$$NF = \frac{B}{A+C}, y_F = \frac{C}{A+C}$$

Leaching Solvent-

R_0 = Mass of Soln. (A+C),

$$x_0 = \frac{C}{A+C},$$

$$N_0 \approx 0.$$



Leached Solution. (R)

R_1 = mass of Soln.

$$x_1 = \frac{\text{mass of C}}{\text{mass of (A+C)}}$$

$$NR_1$$

Leached Solid.

B_1 , Mass of Insoluble.

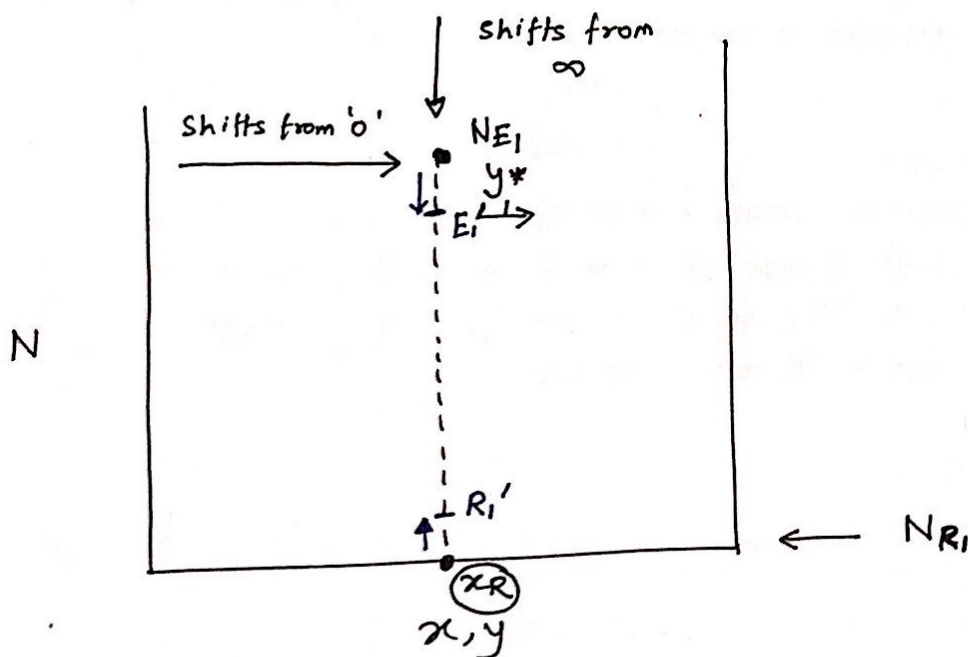
E_1 , mass of A+C

$$N_1 = \frac{B}{E_1} = \frac{B}{(A+C)E_1}$$

$$y_1 = \frac{C \bar{u} E_1}{(A+C)E_1} = \frac{C \bar{u} E}{E_1}$$

- ① We have a ~~sub~~ liquid. Whose composition B, should be ideally zero. ~~or close~~

A = Solvent
C = solute
B = Insoluble.



For the Leached Solid $\rightarrow NE_1$ or $N_1 = ?$ (Ideally ∞),

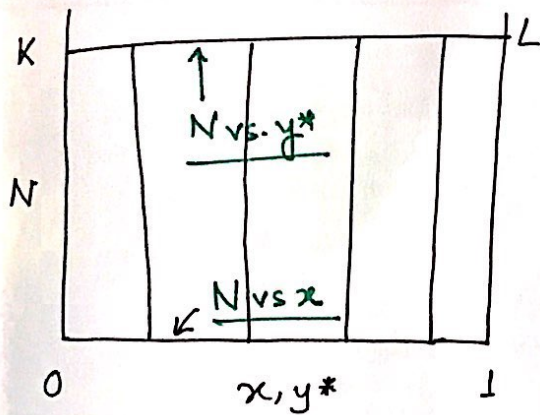
Practical Reality is



← You will have a high conc. Sludge.

Some typical Equilibrium Plots,

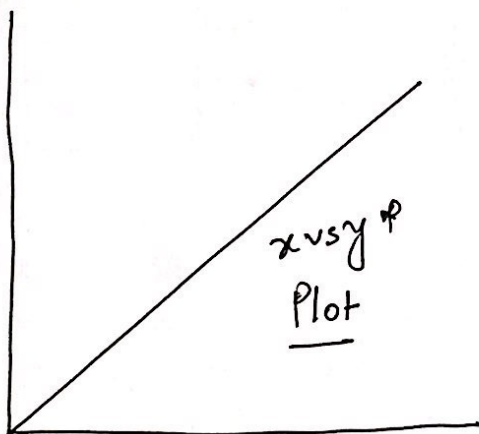
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Vertical tie line,

Here

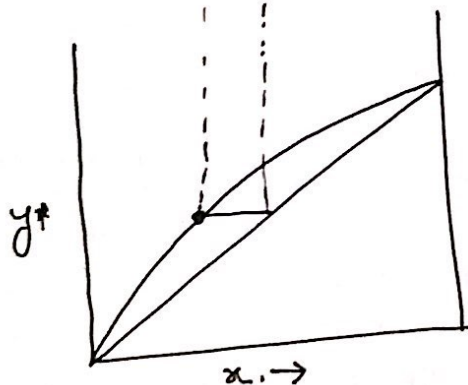
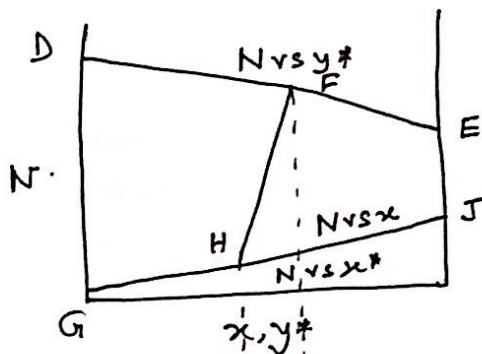
- (1) No adsorption of any Solvent on the Leached Solid.
- (2) No dispersion of any Solid in the leached solution.



The graph indicates that the Solids are settled or drained to the same extent at all Solute Concentration.

This situation is referred to as constant overflow situation.

(2)



For both Cases the Solute has Solubility in the Solvent for all fractions

C is infinitely Soluble in A

* The line GHJ, which provides the composition of the withdrawn solution, lies above $N=0$, which is possible only when either Solid B has a limited Solubility in the Leached solution or, incompletely settled liquid, has been withdrawn.

The tie lines are not vertical.

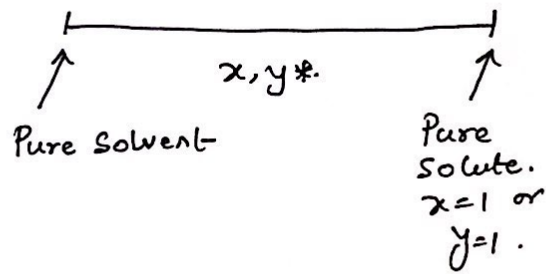
- (1) Insufficient time of Contact for the leaching solvent to dissolve all Solute.

- (2) Preferential adsorption of the Solute.

- (3) Solute is Soluble in the Solid B and distributes unequally into the Solid and liquid phases at equilibrium.

There was a major assumption in both the previous plots,

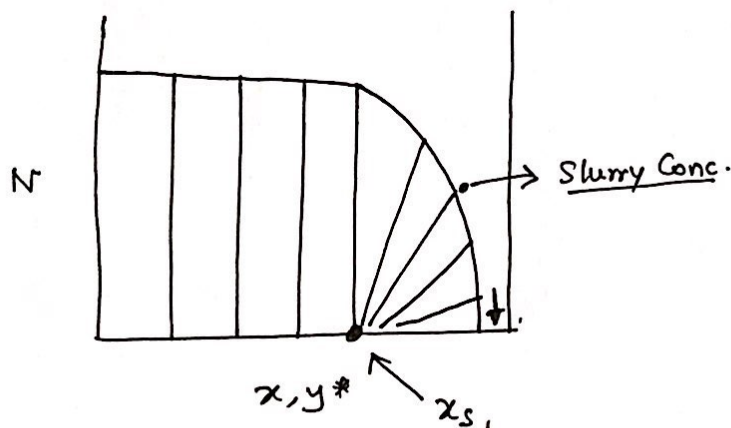
The assumption was that the solute has complete solubility for all values of solvent concentration!



* This will hardly happen.

* So, what is a more likely situation.

There will be a limiting solubility of the solute



Why does slurry conc. start reducing below B

$$N = \frac{B}{A-C}$$

* C. concentration increases.

So in this case, no clear solution stronger than x_s is possible.

Single Stage Leaching Unit

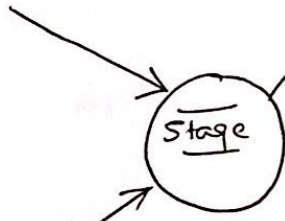
For Leaching

A → Solvent.
B → Insoluble.
C → Solute

Solid to be Leached.
B → Mass of Insolubles.

$$N_F = \frac{B}{A+C}$$

$$Y_F = \frac{C}{A+C}$$



Leached Solid.

B → Mass of Insoluble.

E_1 → mass of (A+C) in Solid Stream.

N_1 → Fraction of B on a B Free Basis.

Y_1 → mass fraction of C on a B Free Basis

Leaching Solvent.

R_0 = mass of Soln. (A+C).

x_0 = Mass fraction of C on B free Basis.

Leach Solution

R_1 → mass of Solution (A+C).

x_1 → mass of C / mass of (A+C)

For Pure Solvent, $\underline{x_0 = 0}$, and $\underline{y_F = 1}$.
 $N_0 = 0$.

Each stage represents the entire operation, including mixing of Solids, with Leaching solvent, ~~and~~ upto mechanical sepⁿ, of resulting insoluble phases.

For all practical purpose, B is insoluble in Solvent and a clear. Liquid Leach Solution is obtained.

$$B = N_F \cdot F = E_1 N_1 \quad \text{--- (1)}$$

Also, a solute balance gives. [C balance]

$$F \cdot y_F + R_0 x_0 = E_1 y_1 + R_1 x_1$$

So, the Solvent (A) is as follows: -

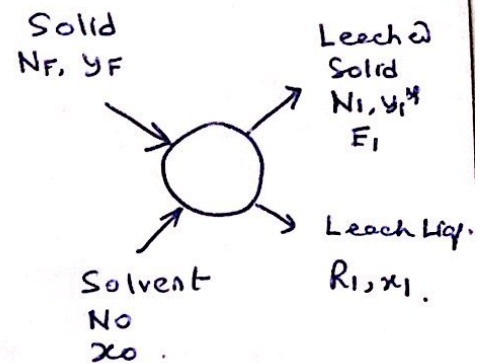
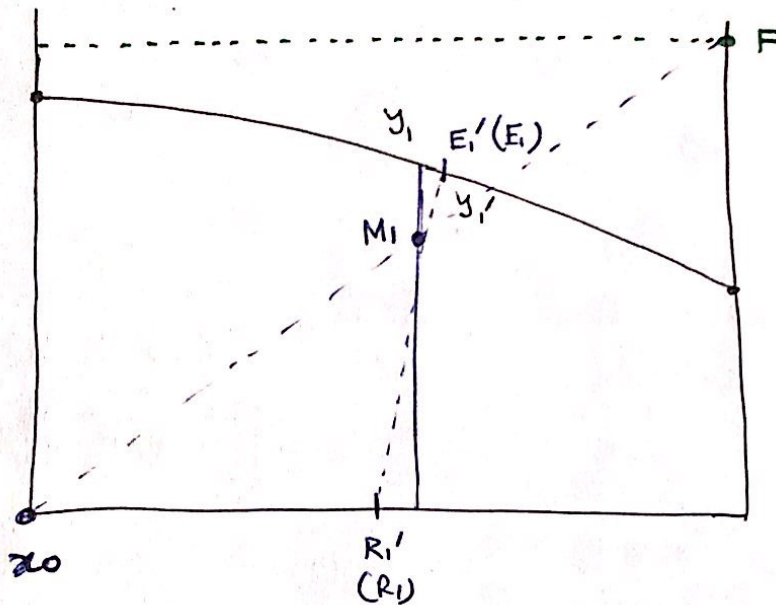
$$F(1 - y_F) + R_0(1 - x_0) = E_1(1 - y_1) + R_1(1 - x_1)$$

Overall Balance (B Free Basis)

$$F + R_0 = E_1 + R_1 + M. \quad - (2)$$

The mixture point on B free Basis is given by

$$N_{M1} = \frac{B}{F + R_0} = \frac{B}{M_1} \quad \text{and} \quad y_{M1} = \frac{y_F F + R_0 x_0}{F + R_0}$$



Once the co-ordinates of the Point E_1 and R_1 are found out, then Equation (1) and (2) to find out E_1 and R_1 .

Stage Efficiency can be calculated.

$$\eta = \frac{y_F - y_1'}{y_F - y_1^o} \quad \left\langle \text{Arguably due to short Contact time} \right\rangle$$

Multi Stage Cross Current Leaching $\rightarrow$.

In this approach the leached Solids of the First stage is contacted with the fresh Solvent.

This process is ideally suited in systems where there is tendency of the solute to remain stuck ~~in the~~ within the porous, insolubles.

Example: 13.2, worked out, Page 749

Caustic Soda is made by treatment of Ca(OH)_2 with solution of Sodium Carbonate. The resultant slurry contains particles of Calcium Carbonate (CaCO_3) suspended in 10% solution of NaOH , 0.125 kg. suspended solid / kg solution. This is settled and the clear sodium hydroxide solution withdrawn and replaced by equal weight of water, and the mixture thoroughly agitated. After repetition of this procedure (a total of two freshwater washes), what fraction of the original NaOH in the slurry remains unrecovered and therefore lost in the sludge? The settling characteristics show adsorption of the solute on the solid.

The Equilibrium data is given as.

x = Wt fraction of NaOH in clear soln.	N = kg CaCO_3 / kg soln. in settled sludge.	y^* = Wt. fraction NaOH in soln. of settled sludge.
0.0900	0.495	0.0917.
0.0700	0.525	0.0762.
0.0473	0.568.	0.0608.
0.0330	0.600	0.0452.
0.0208	0.620	0.0295.
0.01187	0.650	0.0204.
0.00710	0.659	0.01435
0.00450	0.666	0.01015

The equilibrium data and tie line are plotted.

Here B is CaCO_3 .

$A + C$ = NaOH and Water.

10% NaOH in Water.

C = NaOH (Solute)

A = Water.

If we take basis as 1 kg ~~kg~~ of Solution

Then $C = 0.1$ kg

$B = 0.125$ kg.

$A = 0.9$ kg

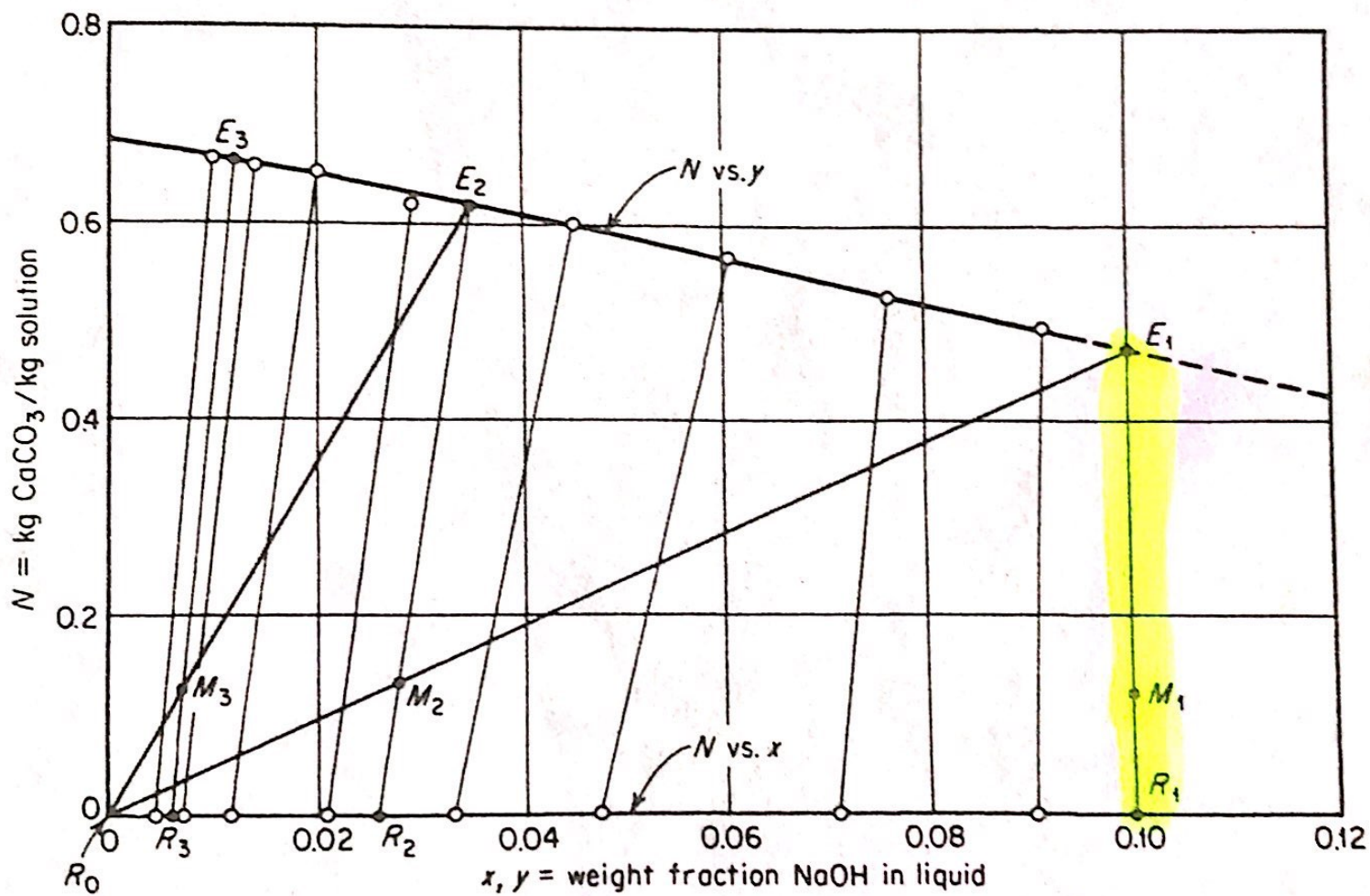


Figure 13.27 Solution to Illustration 13.2.

Basis : 1 kg ~~of~~ Solution in the original mixture.

Which contains 0.1 kg NaOH (C) and 0.9 kg H₂O (A).

$$B = 0.125 \text{ kg CaCO}_3.$$

Mixture Composition. $N_{M_1} = 0.125$
 $y_{M_1} = 0.10$

So First we locate point M, and then draw the tie line through it. So we get the Composition of E₁ and R₁.

From graph. $N_1 = 0.47$
 $y_1 = 0.10$

$$\therefore E_1 = \frac{B}{N_1} = \frac{0.125}{0.47} = 0.266 \text{ kg Solution in the Sludge.}$$

$$\therefore \text{Clear Soln} = 1 - 0.266 = 0.734 \text{ kg Clear soln. withdrawn after the first stage.}$$

Stage 2: $R_0 = 0.734 \text{ kg. water (Solvent) added.}$
 $x_0 = 0.$

$$\therefore M_2 = E_1 + R_0 = (0.266 + 0.734) \text{ kg} = 1.0 \text{ kg.}$$

$$N_{M_2} = \frac{B}{E_1 + R_0} = 0.125.$$

So we join R₀ E₁, where its value equals N_{M₂}, that's point M₂. Get the tie line from there. to get Point E₂

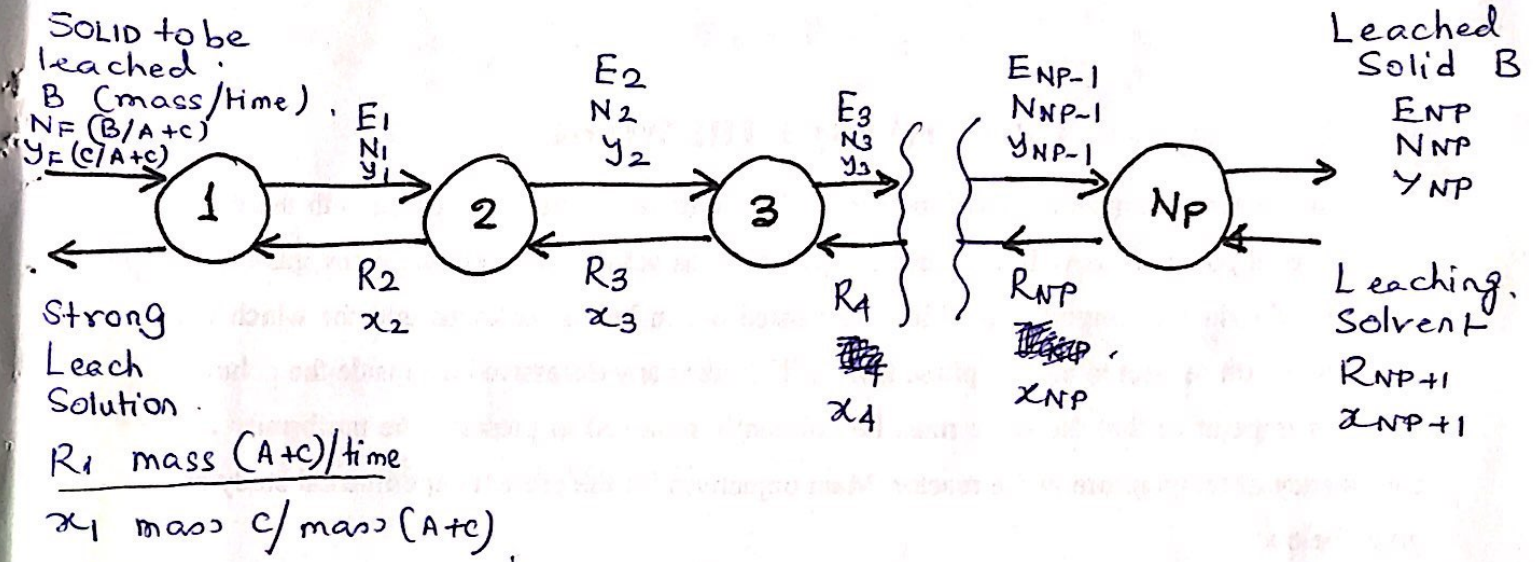
$$N_2 = 0.62, y_2 = 0.035$$

$$E_2 = \frac{B}{N_2} = \frac{0.125}{0.62} = 0.202 \text{ kg.}$$

$$\therefore \text{Solution withdrawn is } 1 - 0.202 = 0.788 \text{ kg.}$$

Similarly for stage 3 find out E₃ and y₃.
 Unrecovered NaOH is (E₃ y₃)

Multi Stage Counter Current Leaching.



In the above flowsheet, it is assumed that solid B is insoluble and is not lost in the clear solution.

A solvent balance for the entire plant is

$$F + R_{NP+1} = R_1 + E_{NP} = M. \quad (3)$$

and solution Balance is $\Rightarrow \boxed{F - R_1 = E_{NP} - R_{NP+1} = M}$

$$F y_F + R_{NP+1} x_{NP+1} = R_1 x_1 + E_{NP} y_{NP} = M y_M.$$

M represents the hypothetical B free mixture obtained by mixing solids to be leached and the leaching solvent.

The co-ordinates of the point M are

$$N_M = \frac{B}{F + R_{NP+1}}$$

$$y_M = \frac{F y_F + R_{NP+1} x_{NP+1}}{F + R_{NP+1}}$$

Similarly one can write. (based on a soln. balance)

$$F - R_1 = E_1 - R_2 = E_2 - E_3 = \Delta R.$$

ΔR represents the constant difference in flow $E - R$.
(usually negative).

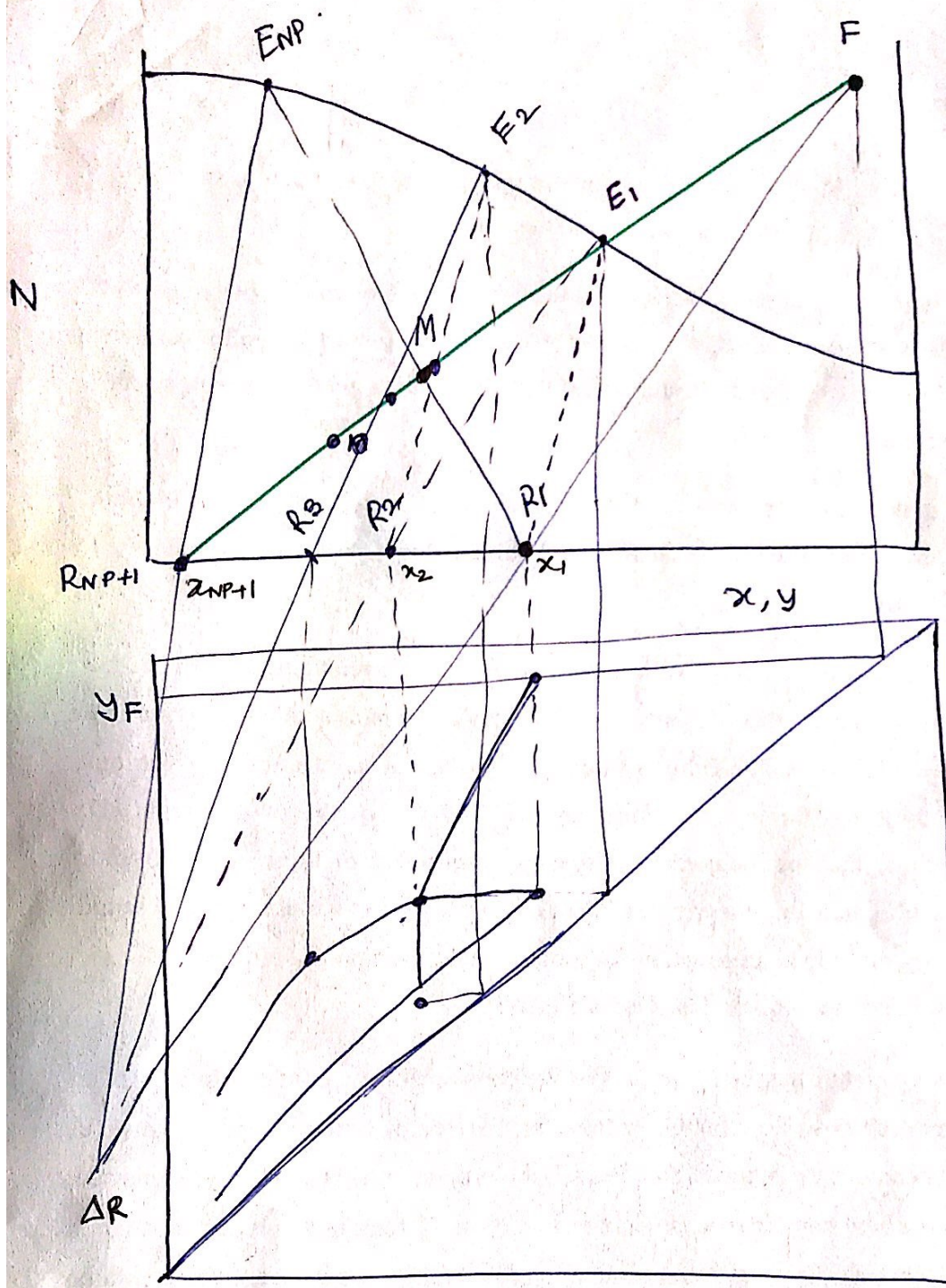
After locating ΔR , a line from E_1 to ΔR provides R_2 and so forth.

==

The points E_{NP} and ~~R_{NP}~~ R_1 represents the effluent from the cascade.

$$F - R_1 = E_{NP} - R_{NP+1} = \Delta R.$$

Similarly $F - R_1 = E_3 - R_4 = \Delta R.$



Locate F
Locate R1
Locate ENP
Locate R_{NP+1}

To do so
We need
 y_F, x_{R1}
 y_{NP}, x_{RNP+1}
and correctly
N.

The points ENP and R1, represents the effluents from the cascade, must lie on a line ~~passing~~ passing, thru' M., and Further ENP will be on the equilibrium Curve

Eqn (3) can be re-written as

$$F - R_1 = ENP - R_{NP+1} = \Delta R$$